

IMO(13th) – 1971

First Day

1. Prove that the following statement is true for $n = 3$ and for $n = 5$, and false for all other $n > 2$: For any natural real numbers $a_1, a_2, \dots, a_n$,

$$(a_1 - a_2)(a_1 - a_3) \dots (a_1 - a_n) + (a_2 - a_1)(a_2 - a_3) \dots (a_2 - a_n) + \dots + (a_n - a_1)(a_n - a_2) \dots (a_n - a_{n-1}) \geq 0.$$
2. Given a polyhedron P_1 with 9 vertices $A_1, A_2, \dots, A_9$, let us denote by $P_2, P_2, \dots, P_9$ the images of P_1 under the translation mapping the vertex A_1 to $A_2, A_3, \dots, A_9$. Prove that among the polyhedral $P_1, P_2, P_2, \dots, P_9$ at least two have a common interior point.
3. Prove that a sequence $2^n - 3$ ($n > 1$) contains a subsequence of numbers relatively prime in pairs.

Second Day

4. Given a tetrahedron ABCD all of whose faces are acute angled triangles, let $\sigma = \angle DAB + \angle BCD - \angle ABC - \angle CDA$. Consider all closed lines XYZTX whose vertices X, Y, Z, T lie in the interior of segments AB, BC, CD, DA respectively. Prove that:
 - (a) If $\sigma \neq 0$, then there is no broken line XYZT of minimal length.
 - (b) If $\sigma = 0$, then there are infinitely many such broken lines of minimal length. That length equals $2AC \sin(\alpha / 2)$, where $\alpha = \angle BAC + \angle CAD + \angle DAB$.
5. Prove that for every natural number $m \geq 1$ there exists a finite set S_m of points in the plane satisfying the following condition: If A is any point in S_m , then there are exactly m points in S_m whose distance to A equals 1.
6. Consider the $n \times n$ array of nonnegative integers:

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

With the following property: If an element a_{ij} is zero, then the sum of the elements of the i th row and j th column is greater than or equal to n . Prove that the sum of all the elements is greater than or equal to $\frac{1}{2}n^2$.

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